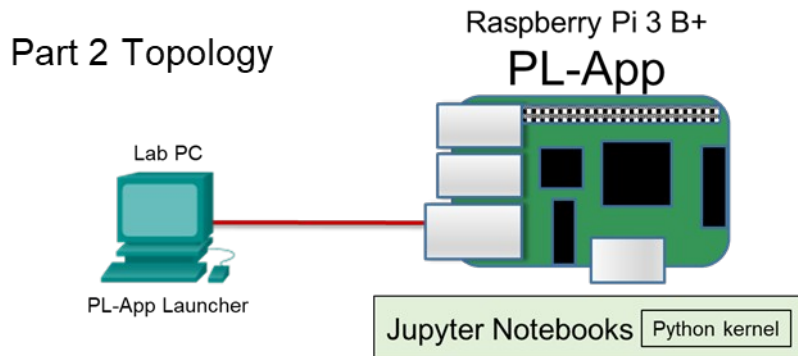


Lab – Set Up the IoT Security Lab - Keep Standard Topology

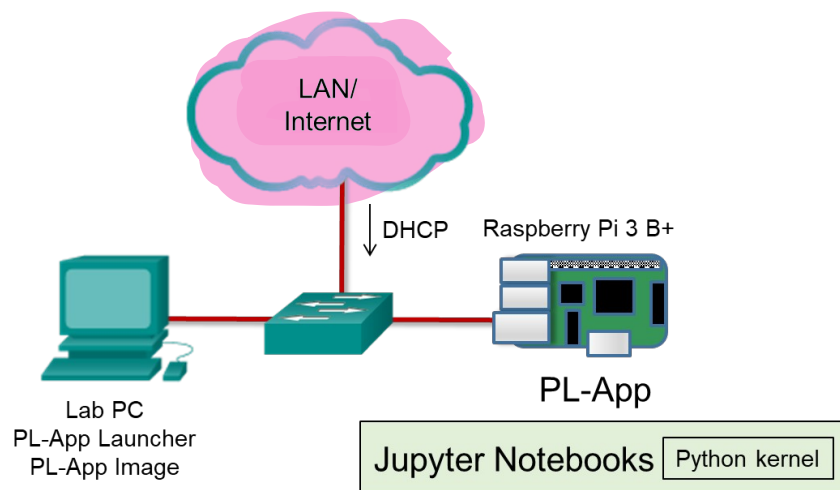
Normal IoT Security Topology – Ethernet Cable required

This is the normal Cisco IoT security topology, which requires a direct Ethernet cable connection from your host computer, on which the Kali VM will be running, to the Raspberry Pi with the PL-App.



To avoid the possibility that Kali could be used on your home network, the Kali VM would be directly connected to the Raspberry Pi in Part 2 of this lab. In addition, Kali would be preconfigured with an address on the 203.0.113.0/24 network, which is reserved for documentation and testing.

Keep Standard Topology



In this document the normal IoT security topology will not be used. Rather the standard topology from the Connecting Things course is kept, such that both the host computer, on which the Kali VM is running, as well as the Raspberry Pi with the PL-App will connect to your home network. This means that the Raspberry Pi as well as your host computer (and the Kali VM) will get an IP address from your home network. This also means that the Kali VM can reach any host of your home network. You are responsible to not misuse the Kali VM!

Objectives

Part 1: Setup the Lab Environment

Part 2: Import the IoT Security Virtual Machines

Background / Scenario

Computing power and resources have increased tremendously over the last 10 years. A benefit of having multicore processors and large amounts of RAM is the ability to use virtualization. With virtualization, one or more virtual computers operate inside one physical computer. Virtual computers that run within physical computers are called virtual machines (VMs). VMs are often called guests, and physical computers are often called hosts. Anyone with a modern computer and operating system can run VMs.

In this lab, you will set up and explore the lab environment that will be used in this course. A VM is used for many of the labs in this course. The VM is created with Oracle VirtualBox or the UTM app. The VM file contains a special version of Linux called Kali. **Kali is a very popular Linux distribution that contains many tools that are used for assessing network security.** VirtualBox (and UTM) allow you to run this version of Linux on a Mac or PC as a VM. You can use this VM to interact with other hosts on your network.

Note: Only use Kali tools on networks on which you are authorized to do so. Abuse of the Kali tools will be a violation of your ethical hacking agreement.

Required Resources

- Host computer with at least 4 GB of RAM and 15 GB of free disk space
- Oracle VirtualBox or the macos UTM app.
- IoT Security Kali Linux OVA . **Do not use the original Kali VM from Cisco!** On the Sciebo server you will find two versions of the Kali VM. Use the Virtualbox image for all host computers other than Mac computers with the Apple M1 chip. For Mac computers with the Apple M1 chip use the UTM VM. **In the version on the Sciebo server set the language and keyboard settings have been changed to German, whereas the original version from Cisco uses US keyboard layout.**
- **Note:** In some labs from the Cisco IoT security course a Metasploitable VM is used. **However, we will not use the Metasploitable VM. So, you need not download the Metasploitable VM!**
- Internet connection
- **An Ethernet patch cable is not required, if you keep the standard lab topology.**

Part 1: Setup the Lab Environment (for Non Apple M1 Chip Computers)

Follow these instructions keeping the standard topology shown in the “Keep Standard Topology” section of this lab. What is important is that you can identify your own target Raspberry Pi using PL-App launcher.

Step 1: Download and Install Oracle VirtualBox.

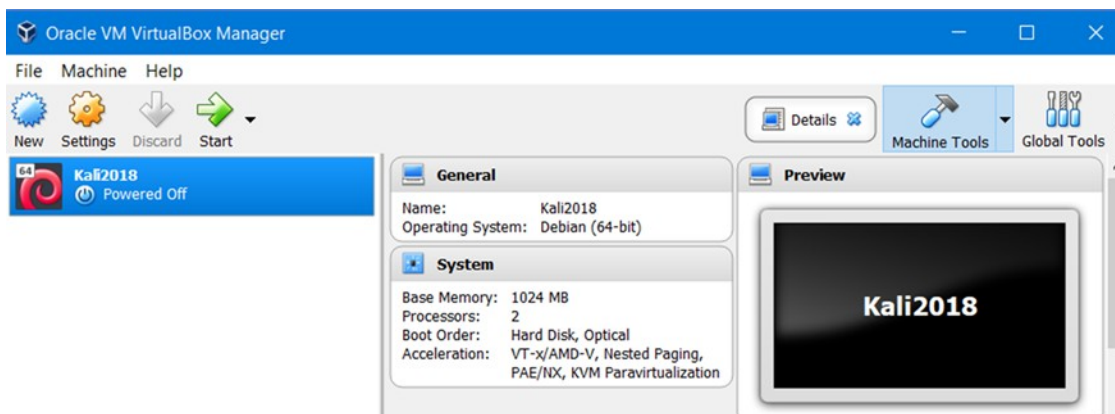
- Navigate to the Oracle VirtualBox downloads [page](#). I
- Choose and download the appropriate installation file for your operating system.
- After you have downloaded the VirtualBox installation file, run the installer and accept the default installation settings.
- Download the IoT Security Virtualbox VM file from Sciebo: [Kali Virtual machines/Virtualbox/IoTSec_Kali.ova](#).

Step 2: Import the Kali virtual machine appliance into VirtualBox.

You will open VirtualBox and import the IoT Security Kali VM .ova file to create a Kali virtual machine.

Note: The screen may look different depending on your version of VirtualBox.

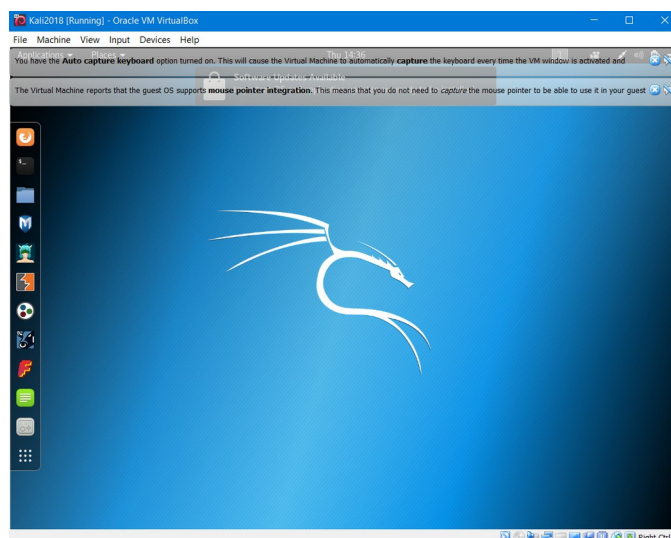
- Open **VirtualBox**. From the file menu, select: **File > Import Appliance**. Locate and select the **Kali_IoTSec.ova** file that you downloaded and click **Next**.
- A new window will appear presenting the settings suggested in the OVA archive. Check the **"Reinitialize the MAC address of all network cards"** box at the bottom of the window. Leave all other settings as default. Click **Import**.
- After the import is complete, VirtualBox will show the new Kali VM in its inventory. Your Kali Linux VM file name might be different than the graphic shown below.



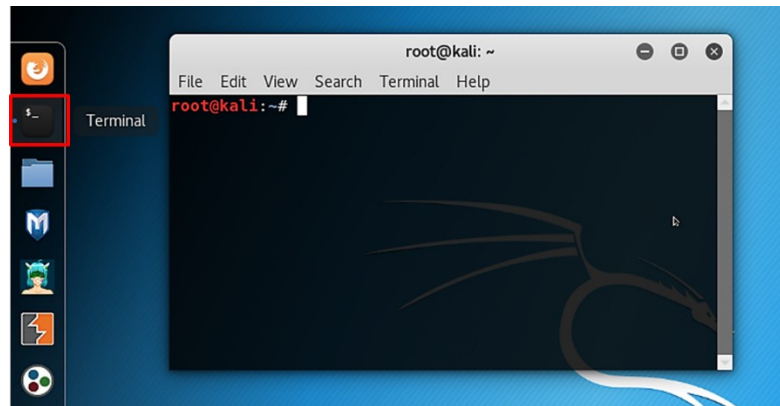
Step 3: Verify Network Connectivity.

In this step, you will ensure that networking is configured correctly between the VM and the Raspberry Pi.

- Return to the VirtualBox VM player window. **Highlight the Kali VM in the list.**
- Click the green **Start** arrow in the menu bar. After a brief delay, you should see a new window open that displays a **Username:** field.
- Enter the username of **root** and click **Next** to continue. Use **toor** for the password and click **Sign in**. If you have typed the username incorrectly for the Kali VM, click **Cancel** to input the correct username.
- After Kali starts, you should see the desktop as shown below.



- e. Click the **Terminal** icon in the desktop applications bar as shown in the figure.



- f. After the script executes, at the terminal prompt, type **ifconfig** or **ip a**:

```
root@kali:~# ifconfig
```

What IP address was assigned to the VM eth0 interface? 192.168.1.151

- g. Minimize the VirtualBox window and open PL-App. Select the **Available Devices** tab. You should see your **Raspberry Pi** listed. Make note of the IP address of your Raspberry Pi.

IP address of the Raspberry Pi: 192.168.1.138

- g. Return to the VM. At a terminal prompt, **ping** the IP address of your Raspberry Pi. Use the **-c 5** parameter to limit your ping to **five echo requests**. If everything is working properly, you should see five successful echo replies. You have now tested the connection between the **Kali Linux virtual machine** and the **Raspberry Pi**.
- h. Now open the **Firefox ESR** browser from the Kali desktop applications bar. Its icon is just above the Terminal icon.
- i. Type the **IP address of the Raspberry Pi** into the address bar and press **Enter**. You should see the **Jupyter notebooks home page** appear.

Note: Because the VM shares the same network adapter as the physical host computer, you can access the Pi from either machine.

Step 4: Shut down the VM.

- a. In the VirtualBox menu in the Kali VM window, select **File > Close**.
- b. Click the **Save the machine state** radio button and click **OK**. The next time you start the virtual machine, you will be able to resume working in the operating system in its current state.

The other two options are:

- **Send the shutdown signal:** This option simulates safely shutting down a physical computer.
- **Power off the machine:** This option simulates an unsafe shutdown of a physical computer.

Part 2: Setup the Lab Environment for Apple M1 Chip Computers only

Follow these instructions keeping the standard topology shown in the “Keep Standard Topology” section of this lab. What is important is that you can identify your own target Raspberry Pi using PL-App launcher.

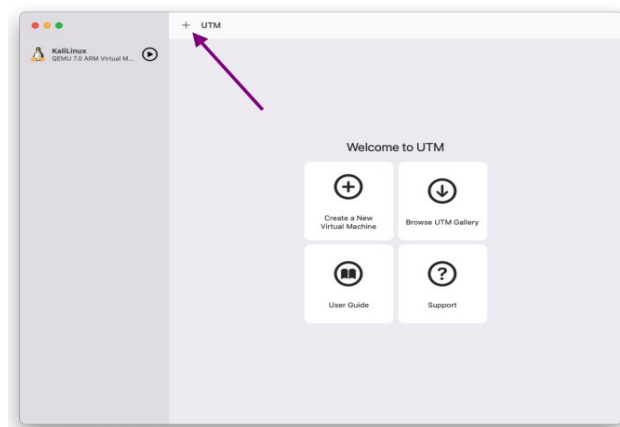
Step 1: Download and Install the UTM APP.

- Download the UTM Mac app version 3.2.1. from Sciebo: [Kali Virtual Machines/UMT/UTM.3.2.1.dmg](#). Mount the DMG file by double clicking it and then drag the UTM app into the /Applications folder. UTM should now be installed.
- Download the IoT Security UTM VM file from Sciebo and save it to a sensible directory on your harddisk: [Kali Virtual machines/UTM/KaliLinuxCiscoIoTSec.utm](#).

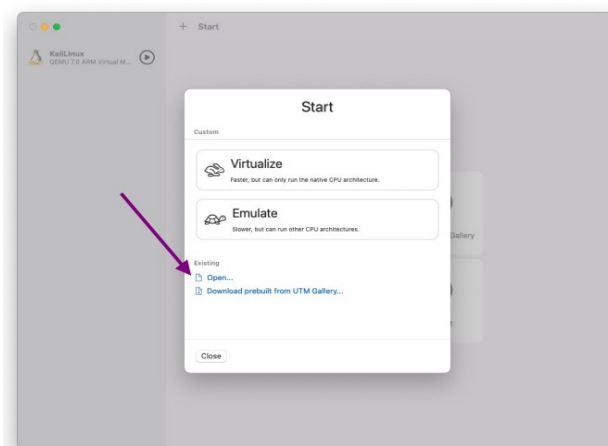
Step 2: Add the Kali virtual machine appliance to UTM.

You will open UTM and add the IoT Security Kali VM .utm file to add the Kali VM to UTM:

- Open the **UTM app**. Click on the **plus symbol** in the top line right left to **UTM**:



Then click on **Open** just beneath **Existing**:

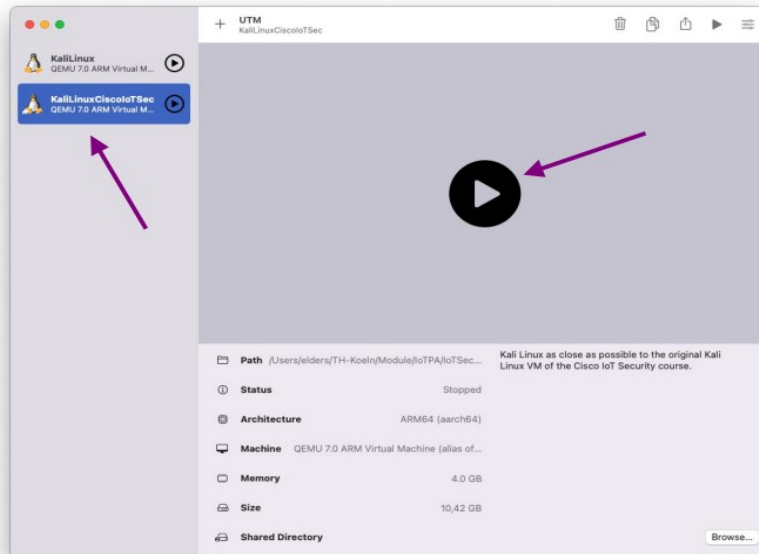


In the following file open file dialog change to the directory, where you saved the .utm file, and click on it.

Step 3: Verify Network Connectivity.

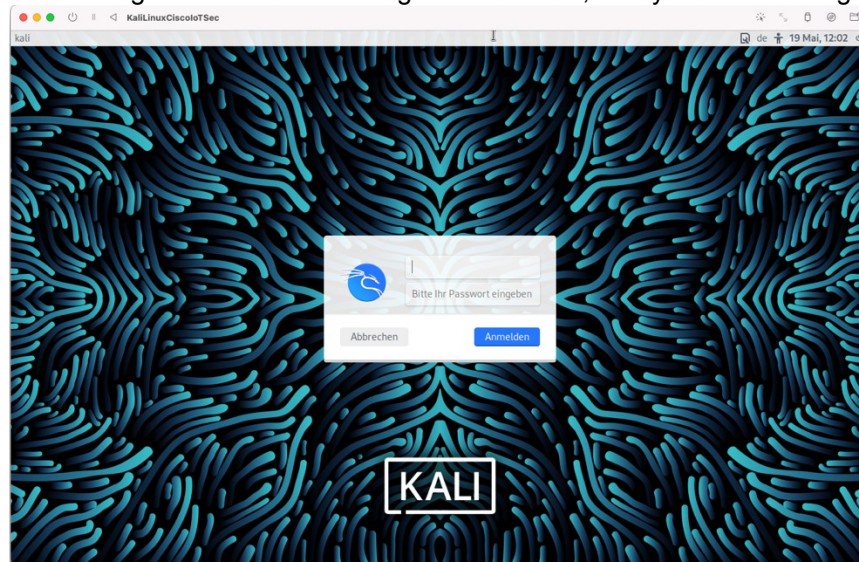
In this step, you will ensure that networking is configured correctly between the Kali VM and the Raspberry Pi.

- a. After import the Kali VM will be show as **KaliLinuxCiscoIoTSec** in the VM pane on the left:



Press the big play button in the VM screen snapshot window to start the VM.

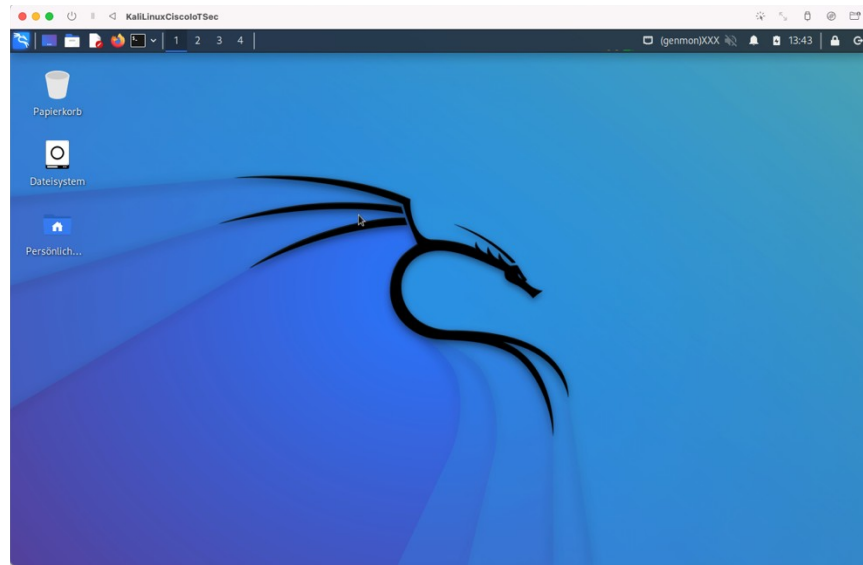
- b. When the VM is starting the screen size changes a few times, finally the Kali VM login screen is shown:



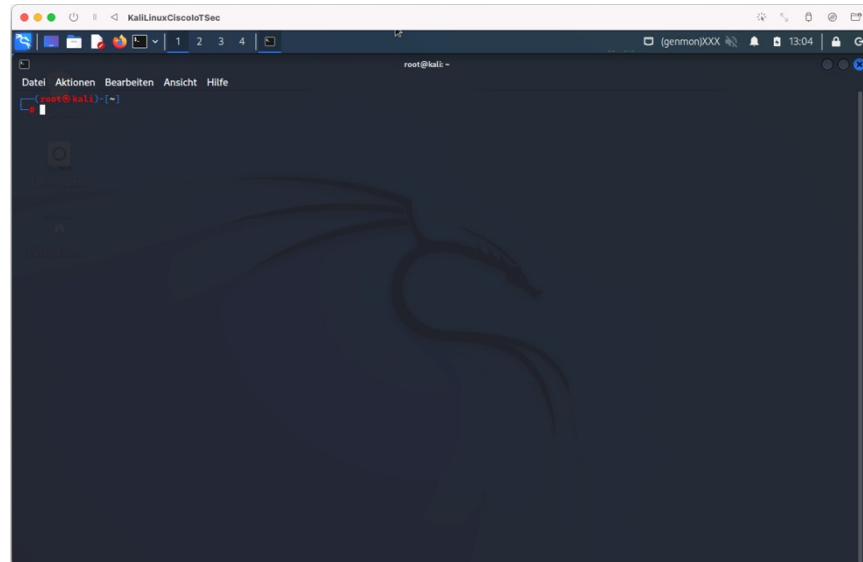
- c. Enter the username of **root** and use **toor** for the password and click **Anmelden**. If you have typed the username incorrectly for the Kali VM, click **Abbrechen** to input the correct username.
- d. **Hint:** The size of the VM window cannot be changed from the host, e.g you cannot resize the VM window by dragging the edges or corners of the window. You can however change the size of the VM window by setting a different screen resolution within the VM. If you want to change the window of the VM click on the **Anwendungen** icon in top left corner of the VM and then select **Einstellungen -> Anzeige** and set the desired screen resolution beneath **Auflösung**. Then the VM window will be resized accordingly.

Lab – Set Up the IoT Security Lab Topology

- e. After Kali login, you should see a similar desktop as below. Perhaps the terminal window is already open and maximized:



- f. If the terminal is not already open, click the **Terminal** icon in the desktop applications bar as shown in the figure to open the terminal:



- g. After the script executes, at the terminal prompt, type **ifconfig** or **ip a**.

```
root@kali:~# ifconfig
```

What IP address was assigned to the VM eth0 interface? _____

- h. Minimize the UTM window and open PL-App. Select the Available Devices tab. You should see your Raspberry Pi listed. Make note of the IP address of your Raspberry Pi.

IP address of the Raspberry Pi: _____

- c. Return to the VM. At a terminal prompt, **ping** the IP address of your Raspberry Pi. Use the **-c 5** parameter to limit your ping to five echo requests. If everything is working properly, you should see five successful echo replies. You have now tested the connection between the Kali Linux virtual machine and the Raspberry Pi.

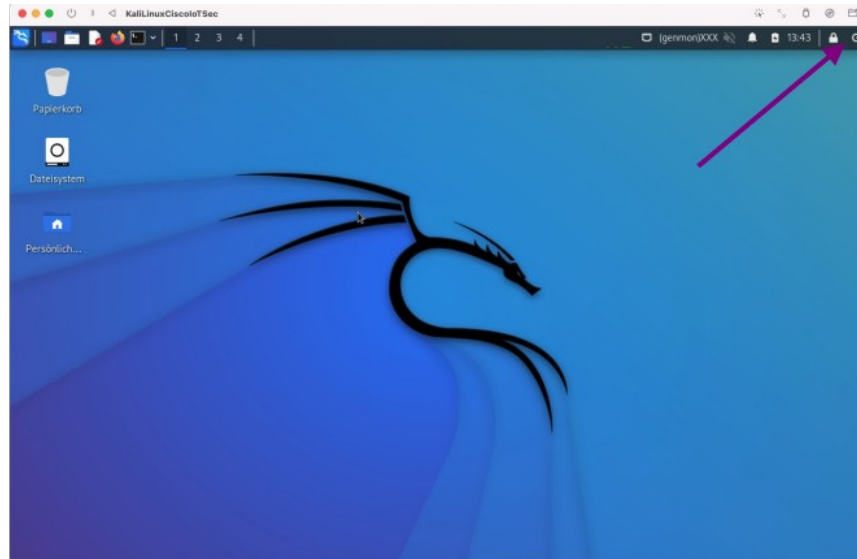
Lab – Set Up the IoT Security Lab Topology

- d. Now open the **Firefox ESR** browser from the Kali desktop applications bar. Its icon is just above the Terminal icon.
- e. Type the IP address of the Raspberry Pi into the address bar and press Enter. You should see the Jupyter notebooks home page appear.

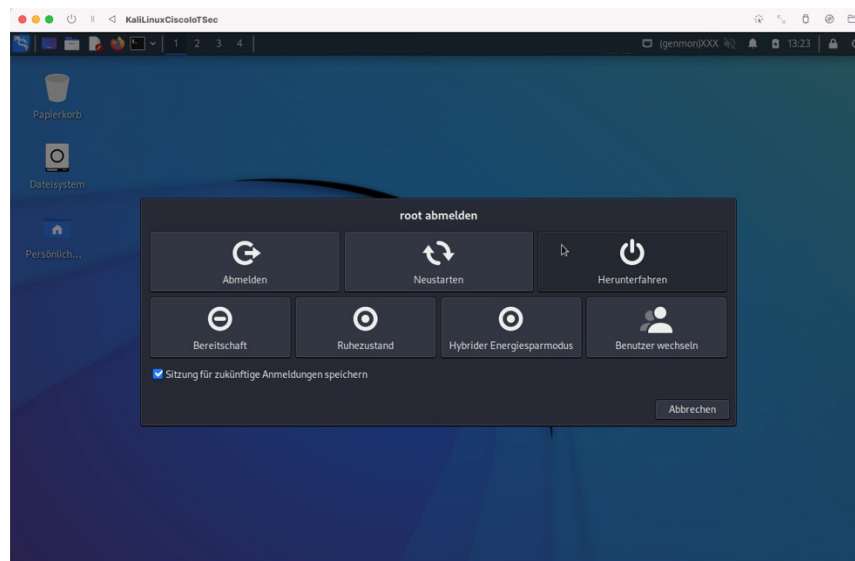
Note: Because the VM shares the same network adapter as the physical computer, you can access the Pi from either machine.

Step 4: Shut down the VM.

- a. Start the shutdown dialog of the Kali VM by clicking on the button in the upper right corner:



- b. Select **Herunterfahren** to shut down the VM:



Selecting the **Save the machine state / Sitzung für zukünftige Anmeldungen speichern** radio button will preserve the state of the VM so that you will be able to resume working in the operating system in its current state the next time you start the VM.